$f \in \mathcal{L}_2(-\infty, \infty)$ Bilateral Laplace transform: $\int_{-\infty}^{\infty} f(t) e^{-st} dt =: F(s) = \Lambda(f(t))$

$\Lambda^{-1}(F(s)) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(s) e^{st} dt$

$$\Lambda : \mathcal{L}_2(-\infty, \infty) \longmapsto \mathcal{L}_2(j\mathbb{R})$$

s.t. $\Lambda(f(t)) := \int_{-\infty}^{\infty} f(t) e^{-st} dt.$

Λ is a isometric isomorphism.

$$\langle f(t), g(t) \rangle = \langle \Lambda(f(t)), \Lambda(g(t)) \rangle = \langle F(j\omega), G(j\omega) \rangle = \langle F(s), G(s) \rangle$$

$$\boxed{\mathcal{L}_2(-\infty, \infty) \cong \mathcal{L}_2(j\mathbb{R})}$$

$$\boxed{\mathcal{L}_2[0, \infty) \cong \mathcal{H}_2} \qquad \boxed{\mathcal{L}_2(-\infty, 0] \cong \mathcal{H}_2^{\perp}}$$

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$$\Lambda(f(t)) := \int_{-\infty}^{\infty} f(t) e^{-st} dt =: F(s) \in \mathcal{L}_2(j\mathbb{R}) \leftarrow \text{Fourier transform}$$

$$\mathcal{F}(f(t)) := \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt =: F(j\omega) \in \mathcal{L}_2(j\mathbb{R})$$

$$\downarrow$$

$$\mathcal{F} : \mathcal{L}_2(-\infty, \infty) \longrightarrow \mathcal{L}_2(j\mathbb{R}) \qquad \text{Isometric isomorphism (I.I.)}$$

$$\left.\begin{aligned} &\Lambda_+ : \mathcal{L}_2[0, \infty) \longrightarrow \mathcal{H}_2 \\ &\Lambda(f(t)) := \int_0^{\infty} f(t) e^{-st} dt =: F(s) \in \mathcal{H}_2 \quad \forall f(t) \in \mathcal{L}_2[0, \infty) \end{aligned}\right\} \text{I.I}$$

$$\left.\begin{aligned} &\Lambda_- : \mathcal{L}_2(-\infty, 0] \longrightarrow \mathcal{H}_2^{\perp} \\ &\Lambda(f(t)) := \int_{-\infty}^{0} f(t) e^{-st} dt =: F(s) \in \mathcal{H}_2^{\perp} \quad \forall f(t) \in \mathcal{L}_2(-\infty, 0] \end{aligned}\right\} \text{I.I.}$$

$$\boxed{\mathcal{L}_2 = \mathcal{L}_{2+} \oplus \mathcal{L}_{2-}} \qquad \boxed{\mathcal{L}_2(j\mathbb{R}) = \mathcal{H}_2 \oplus \mathcal{H}_2^{\perp}}$$

$P : \mathcal{V} \longrightarrow \mathcal{V} \qquad \mathcal{V} = \mathcal{X}_1 \oplus \mathcal{X}_2$

$Pv = x_1$ where $x_1 \in \mathcal{X}_1$ and $v = x_1 + x_2$

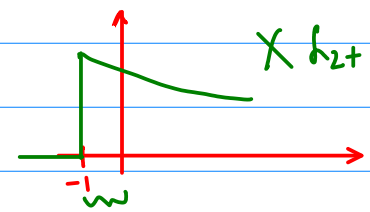
Projectors: $P_+ : \mathcal{L}_2(-\infty, \infty) \longrightarrow \mathcal{L}_2[0, \infty) \qquad P_- : \mathcal{L}_2 \longrightarrow \mathcal{L}_{2-}$

such that $P_+(f(t)) = \begin{cases} f(t), & t \geq 0 \\ 0, & t < 0 \end{cases}$ s.t. $P_-(f(t)) = \begin{cases} 0, & t > 0 \\ f(t), & t \leq 0 \end{cases}$

$$\begin{tikzcd}
\mathcal{L}_{2+} \arrow[r, shift left, "\text{Laplace transf}"] & \mathcal{H}_2 \arrow[l, shift left, "\text{Inv. transf.}"] \\
\mathcal{L}_2(-\infty,\infty) \arrow[u, "P_+"] \arrow[d, "P_-"'] \arrow[r, shift left, "\mathcal{F}"] & \mathcal{L}_2(j\mathbb{R}) \arrow[u, "P_+"'] \arrow[d, "P_-"] \arrow[l, shift left, "\mathcal{F}^{-1}"] \\
\mathcal{L}_{2-} \arrow[r, shift left, "\text{Laplace trans}"] & \mathcal{H}_2^{\perp} \arrow[l, shift left, "\text{Inv. L.T.}"]
\end{tikzcd}$$

e.g. $\dfrac{e^s}{s+1}$

$$\mathcal{L}^{-1}\left(\frac{e^s}{s+1}\right) = \begin{cases} e^{-(t+1)}, & t \geq -1 \\ 0, & t < -1 \end{cases}$$



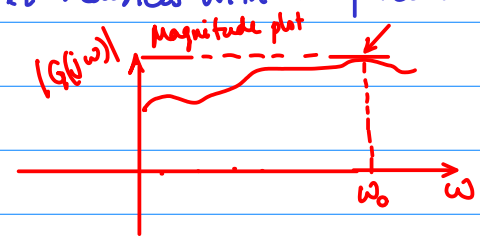
$\mathcal{L}_\infty(j\mathbb{R})$ or $\mathcal{L}_\infty$ is a Banach space of matrix-valued f^n that are essentially bounded on $j\mathbb{R}$ with norm

$$\|F\|_\infty := \operatorname*{ess\,sup}_{\omega \in \mathbb{R}} \sigma_{max} |F(j\omega)|$$

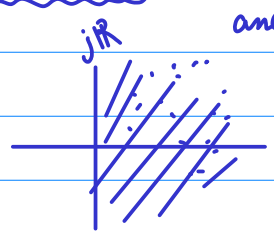
$R\mathcal{L}_\infty \rightarrow$ consists of all rational proper transfer matrices with no poles on $j\mathbb{R}$.

e.g. $G(s) = 1/s+1$

$$\|G\|_\infty = \operatorname*{ess\,sup}_{\omega} \left(1/\sqrt{1+\omega^2}\right)$$



H_∞ space: H_∞ is a (closed) subspace in $\mathcal{L}_\infty$ with f^n that are analytic in $\mathbb{C}^+$ and bounded on the imaginary axis. The H_∞ norm is defined by



$$\|F\|_{H_\infty} := \sup_{Re\{s\}>0} \sigma_{max}[F(s)] = \operatorname*{ess\,sup}_{\omega \in \mathbb{R}} \sigma_{max}[F(j\omega)]$$

RH_∞ : all real, proper rational stable matrices

H_∞^- space: H_∞^- is a (closed) subspace in $\mathcal{L}_\infty$ with f^n that are analytic in $\mathbb{C}^-$ and bounded on the imaginary axis. The H_∞^- norm is defined by

$$\|F\|_{H_\infty^-} := \sup_{Re\{s\}<0} \sigma_{max}[F(s)] = \operatorname*{ess\,sup}_{\omega \in \mathbb{R}} \sigma_{max}[F(j\omega)]$$

RH_∞^- : all real, proper rational antistable matrices $\longrightarrow$ anti-causal matrices / anti-stable matrices

$$\mathcal{L}_\infty(j\mathbb{R}) \not\cong \mathcal{L}_\infty(-\infty,\infty)$$

Properties of $\mathcal{L}_\infty$ and $\mathcal{H}_\infty$ functions:

(I) If $G(s) \in \mathcal{L}_\infty$, then $G(s)\mathcal{L}_2 := \{ G(s)f(s) \mid f(s) \in \mathcal{L}_2 \} \subset \mathcal{L}_2$

(II) If $G(s) \in \mathcal{H}_\infty$, then $G(s)\mathcal{H}_2 := \{ G(s)f(s) \mid f(s) \in \mathcal{H}_2 \} \subset \mathcal{H}_2$

(III) If $G(s) \in \mathcal{H}_\infty^-$, then $G(s)\mathcal{H}_2^\perp := \{ G(s)f(s) \mid f(s) \in \mathcal{H}_2^\perp \} \subset \mathcal{H}_2^\perp$

$\square$ Every function $\underset{\downarrow\, p\times q \text{ matrix}}{G(s)} \in \mathcal{L}_\infty(j\mathbb{R})$ defines a bounded linear operator

$$\mathcal{M}_G : \mathcal{L}_2^q \longrightarrow \mathcal{L}_2^p$$

$$\mathcal{M}_G f := G(s) f(s)$$

$$\|\mathcal{M}_G\| = \|G\|_\infty$$

$$\begin{tikzcd}
\mathcal{L}_2(-\infty,\infty) \arrow[r, "\mathcal{F}^{-1}\mathcal{M}_G\mathcal{F}"] \arrow[d, "\mathcal{F}"'] & \mathcal{L}_2(-\infty,\infty) \arrow[d, "\mathcal{F}"] \\
\mathcal{L}_2(j\mathbb{R}) \arrow[r, "G(s)"] & \mathcal{L}_2(j\mathbb{R})
\end{tikzcd}$$